

coil and material as well as quality of utensils.

Safety tests

Safety tests—that is, insulation resistance, electrical strength and protection against electric shock—were conducted as per relevant Indian standards applicable to appliances.

All the cookers passed safety tests as they were rightly insulated, withstood the electrical strength and had protection against electric shock.

Endurance test at rated voltage (48 hours continuous at maximum power). The appliances were operated at maximum power setting with potable water used as the test load, for a continuous period of 48 hours. The samples were visually checked for any sign of deterioration and breakdown as well as proper functioning after the test.

All the cookers showed no damage.

General and physical tests

As much as performance and safety considerations, the design and overall look of the product also matter to the consumer while buying a product. The appliance should not have an odd shape or design, or sharp edges that may cause any inconvenience to the user. All the cookers were rated on the basis of their design as well as ease of use and convenience.

To assess convenience and ease of use, the tested products were operated through built-in six to eight soft-press switches, where a selection of recipes as well as temperature and time were given. One of the cookers tested had only one soft-touch system, which made it convenient to handle. In two brands, there were two-three soft-press switches, which made selection easier. These cookers were rated higher than the others.

Length of cord and power plug. Most cookers came with a power cable measuring up to 1.25 metres, except for Westinghouse and Prestige, which had a 1.5m long cable. Since all the tested induction cookers, except one (2100 watts), were rated between 1800 watts

TABLE III
Power Consumption of Tested Induction Cookers Versus a LPG Stove Using Base Reference of a Specific Food Item (Water)

(Operating cost in Rs per day for using 10 times a day)

Brand	Per day*	Per month	Per year	Overspend in a year w.r.t. Westinghouse#	Annual difference b/w induction cookers and a LPG stove @ Rs 6.76 per day**
Westinghouse	5.94	178.2	2138.4	—	-295.2
Inalsa	6.24	187.2	2246.4	108	-187.2
Prestige	6.48	194.4	2332.8	194.4	-100.8
Sunflame	6.57	197.1	2365.2	226.8	-68.4
Kenwood	7.02	210.6	2527.2	388.8	+93.6
Khaitan	7.02	210.6	2527.2	388.8	+93.6
Morphy Richards	7.2	216	2592	453.6	+158.4
Philips	7.26	217.8	2613.6	475.2	+180
Havells	7.26	217.8	2613.6	475.2	+180
Bajaj	7.26	217.8	2613.6	475.2	+180
Glen	7.62	228.6	2743.2	604.8	+309.6
Usha	7.62	228.6	2743.2	604.8	+309.6

#Reference taken for calculation w.r.t. the table of water and milk boiling time and actual energy consumed

*Calculated based on the actual consumed in the time taken w.r.t. Westinghouse as reference

**Calculated based on an LPG cylinder cost @ Rs 600 and usage equal to the induction cookers' usage. Gas consumption @ 16.1 g/litre of water boiling in 5.36 minutes calculated @ Rs 6.76/day

and 2000 watts, their power plug too was expected to be of higher rating. Kenwood, Westinghouse, Prestige, Havells and Usha cookers had 15-amp power plugs, while the rest all had 5-amp plugs, which may not be safe for longer use in the case of induction cookers with 2000-watt rating.

For the purpose of overall rating, *Consumer Voice* considered a 1.5m long cord for full score as it provided 25 cm of extra cable, making it convenient to use. Prestige and Westinghouse cookers had a longer power cable of about 1.5 metres.

Dimensions. The diameter of the circle marked on the top plate was measured to find out the effective hot-spot area. In the case of two or more concentric circles, the diameter of the outermost circle was reported. These

circles enable users to use utensils on the effective surface area. Larger appliances may be used for cooking a larger quantity of food. Accordingly, the appliances were rated.

Net weight.

The total weight of an induction cooker indicates its robustness, for it to withstand the utensil's weight as well as the food mass (load) in the utensil. A higher total weight was considered to be better. Havells cooker (2272 gm) was the heaviest, followed by Prestige (2199 gm) and Kenwood (2166 gm). The lightest cookers were Khaitan (1654 gm) and Glen (1754 gm).

Markings. All the cookers provided adequate marking information, except for Kenwood, Khaitan and Prestige, which did not provide technical information. Sunflame did not give details of the kind of utensils to use.

The verdict

The test results matched the hypothesis. The contemporary induction cooker can easily replace a gas burner as it can do all that the gas burner does, and in a more efficient, economical and eco-friendly manner. If the recommended type of steel utensils (ferro-magnetic) with a flat bottom is used, the induction cooker performs much better than any traditional type of stove. The testing team used three different utensils to conduct five tests and found the results of all induction cookers quite satisfactory.